RV College of Engineering®

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SDG9 EL	CS Cluster		
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Project Title	Minimizing Energy Consumption and Delivery Time while Ensuring High Reliability in UAVs		
Under taken at	RV College of Engineering		

SYNOPSIS

Introduction:

Drones are now used to deliver parcels and medicines, support search and rescue, spray crops, inspect industrial assets and watch over large sites. On every one of these missions the operator has to balance three things: how much battery the trip uses, how long it takes, and how likely it is to finish without a failure. Small multirotors fly for only tens of minutes on a charge, so this balance matters. Flight speed is the one setting that affects all three. Flying too slowly wastes energy because the drone is close to hovering, and flying too fast increases drag and stress. Today the speed is usually picked by rule of thumb. In this project we treat the choice of speed as a computational problem: we model the drone mathematically, simulate missions, and use optimisation algorithms to find the best speed that still meets the battery and reliability limits. The project supports SDG 9 (Industry, Innovation and Infrastructure) by making drone-based delivery and emergency response more efficient and dependable.

Objectives:

1. Model the energy, delivery time and reliability of a UAV mission as functions of flight speed, payload, distance and wind.
2. Formulate the choice of speed as a constrained multi-objective problem with battery, reliability and speed limits.
3. Implement and compare grid search, a weighted-sum method and NSGA-II for finding the optimal feasible speed.
4. Build an industry benchmark of commercial drones so the model is set up with real published specifications.
5. Develop a web application where a planner can change the mission and see the recommended speed and trade-offs.

Methodology:

For energy we use the rotary-wing power model of Zeng, Xu and Zhang (2019) and scale it to the mass of each drone category. Delivery time is the distance divided by the ground speed, which includes headwind. Mission reliability is modelled as $R = \exp(-\lambda(V)\cdot T)$, where the failure rate $\lambda(V)$ grows with speed; since no drone company publishes failure rates, its parameters are clearly marked as assumptions. The problem is: minimise energy and time, subject to reliability ≥ 0.995, energy within the usable battery, and speed below the drone's maximum. We solve it with grid search as an exact baseline, a SciPy weighted-sum method, and NSGA-II (pymoo), which returns the whole energy-time Pareto front.

Inputs for six categories (delivery, disaster relief, agriculture, surveillance, mapping and inspection) come from our benchmark. We first checked our code against the Phase-1 reference curve, where the largest error is 0.05 kJ, and all 19 automated tests pass. We then compare the three optimisers, study how payload, headwind and distance change the optimal speed, and compare the result with flying at a fixed 10 m/s. In our current results the optimal speeds save 18-47% energy and 16-60% time compared with 10 m/s. The model and optimisers are served through a Flask web application with an interactive front end.

Data set used:

A UAV Industry Benchmark that we compiled from official specification sheets of 50 commercial drones in six categories, with a source link for every entry. Blank cells mean a specification is not published. Power-model coefficients are taken from Zeng et al. (2019). For validation in Phase-3 we plan to use a public multirotor flight-energy dataset; the exact dataset may change once we check that it matches our model inputs.

Software Requirements:

Python with NumPy and SciPy for the model, pymoo for NSGA-II, Matplotlib for result figures, and pytest for testing. Flask for the web server, with HTML, CSS, JavaScript and Plotly.js for the interface. Microsoft Excel and openpyxl for the benchmark. These choices can be refined during implementation.

Hardware Requirements:

A laptop or desktop is enough to run the simulations, the optimisers and the web application. The study is simulation-based, so no drone hardware or GPU is needed.

Contribution to the field:

The project optimises energy, delivery time and reliability together, where most earlier work looks at energy or time alone. It grounds the model in a sourced benchmark of real commercial drones instead of one assumed aircraft, and it compares an exact method, a weighted-sum method and NSGA-II on the same problem. The web application turns the results into a tool that a delivery or disaster-response planner can actually use to choose a speed.